

Restaurant Recommendation Engine

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1. Project Intuition

Yelp, as one of the most predominant food apps that we use to find food, only shows the average star from other users. Our project presents a customizable restaurant recommendation system that combines structured business data with natural language sentiment analysis to provide highly personalized dining suggestions. We went beyond simple star ratings to understand what truly matters to users.

Data Source: Yelp Open Dataset (50000+ restaurants with 7M+ reviews)
Output: Top 10 restaurants recommended by the model, each with a flashcard



2. Features Engineering

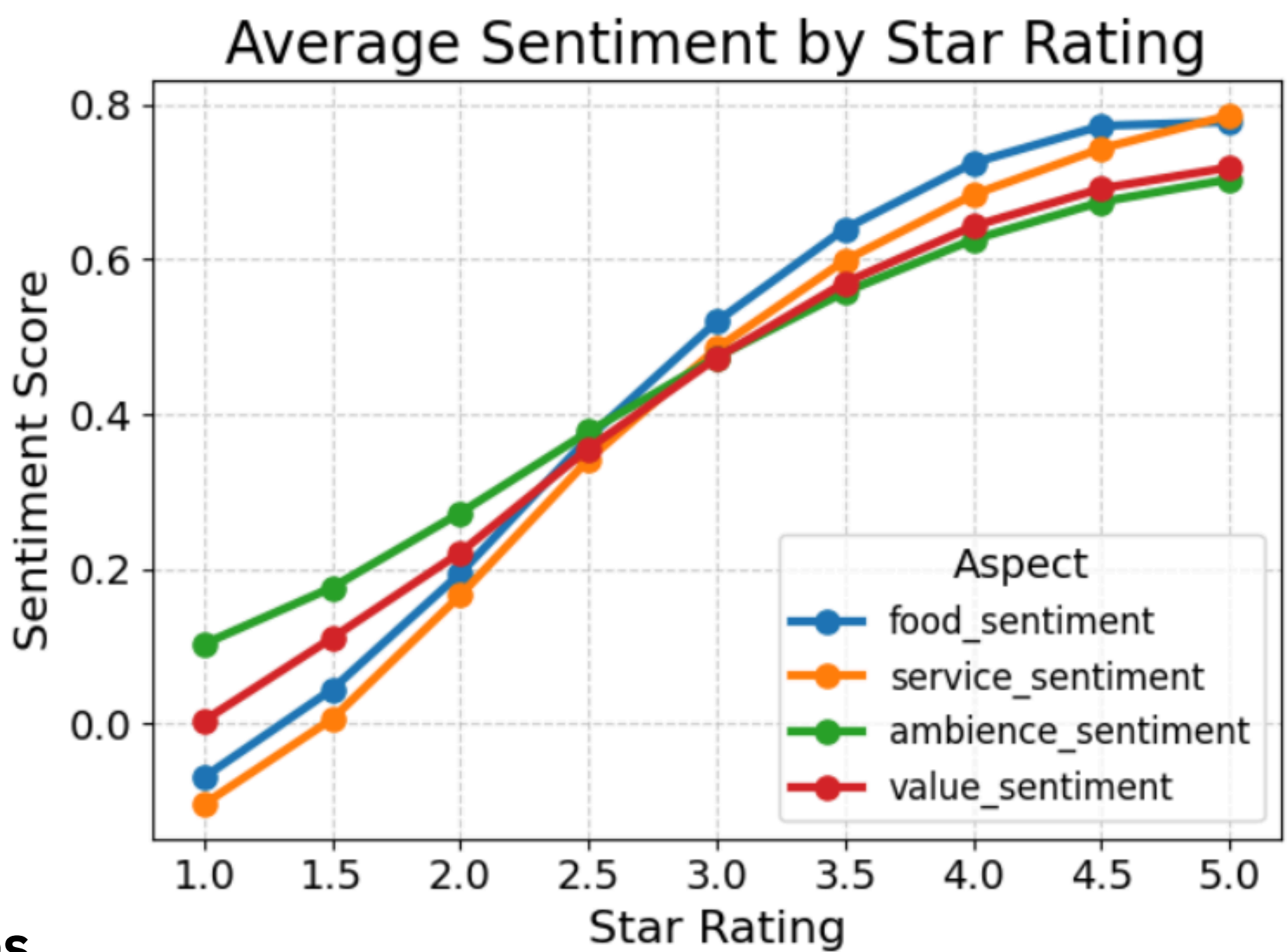
Data Processing

- Remove non-US, non-restaurant business
- Get most frequent attributes (Parking, Price Level...)
- Group restaurant data with reviews
- Extract opening days and hours

Sentiment Analysis

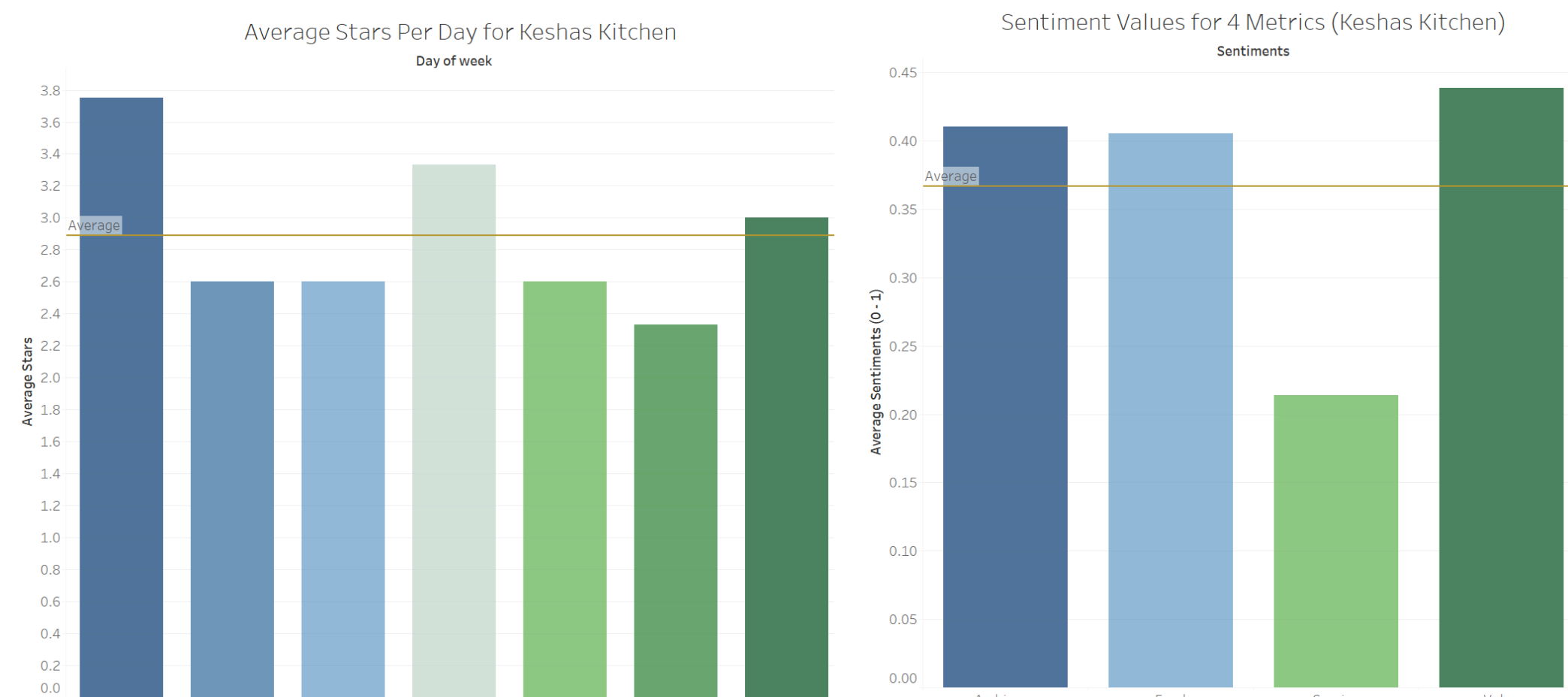
- We analyzed reviews using natural language processing to extract four key sentiment dimensions for each restaurant: food, service, ambience, and value. Each sentiment value ranges from -1 (very negative) to 1 (very positive) and is calculated from pretrained language models.
- Why it matters:** The star associated with a review can depend on different aspects. Sometimes, a user rate a restaurant only by its ambience because he chose the restaurant for an event. On other occasions, exceptional service quality may compensate for otherwise unremarkable food offerings, or vice versa. E.g. A 4.5-star review saying “The steak was so good, but we waited for very long” will result in a high food sentiment score but low service score.

	food_sentiment	service_sentiment	ambience_sentiment	value_sentiment
count	56023	56023	56023	56023
mean	0.612	0.578	0.550	0.554
std	0.222	0.217	0.152	0.179
min	-0.703	-0.868	-0.394	-0.533
25%	0.508	0.486	0.475	0.474
50%	0.672	0.640	0.587	0.605
75%	0.772	0.728	0.655	0.678
max	0.992	0.992	0.977	0.986



We can see the statistics of each sentiments, as well as the increasing trend of average sentiments as the restaurant gets higher rating.

Features



Above are the plots for **Keshas Kitchen**, showing the average stars it received for every day of the week and the different scores it got on each sentiment metric. This latter graph is used in the case that a user identifies this restaurant and wishes to see where the review came from. If a user prioritizes service over ambience, seeing a review broken down via sentiment analysis will help a user decide which restaurant to dine at, even if they have received the same star ratings.

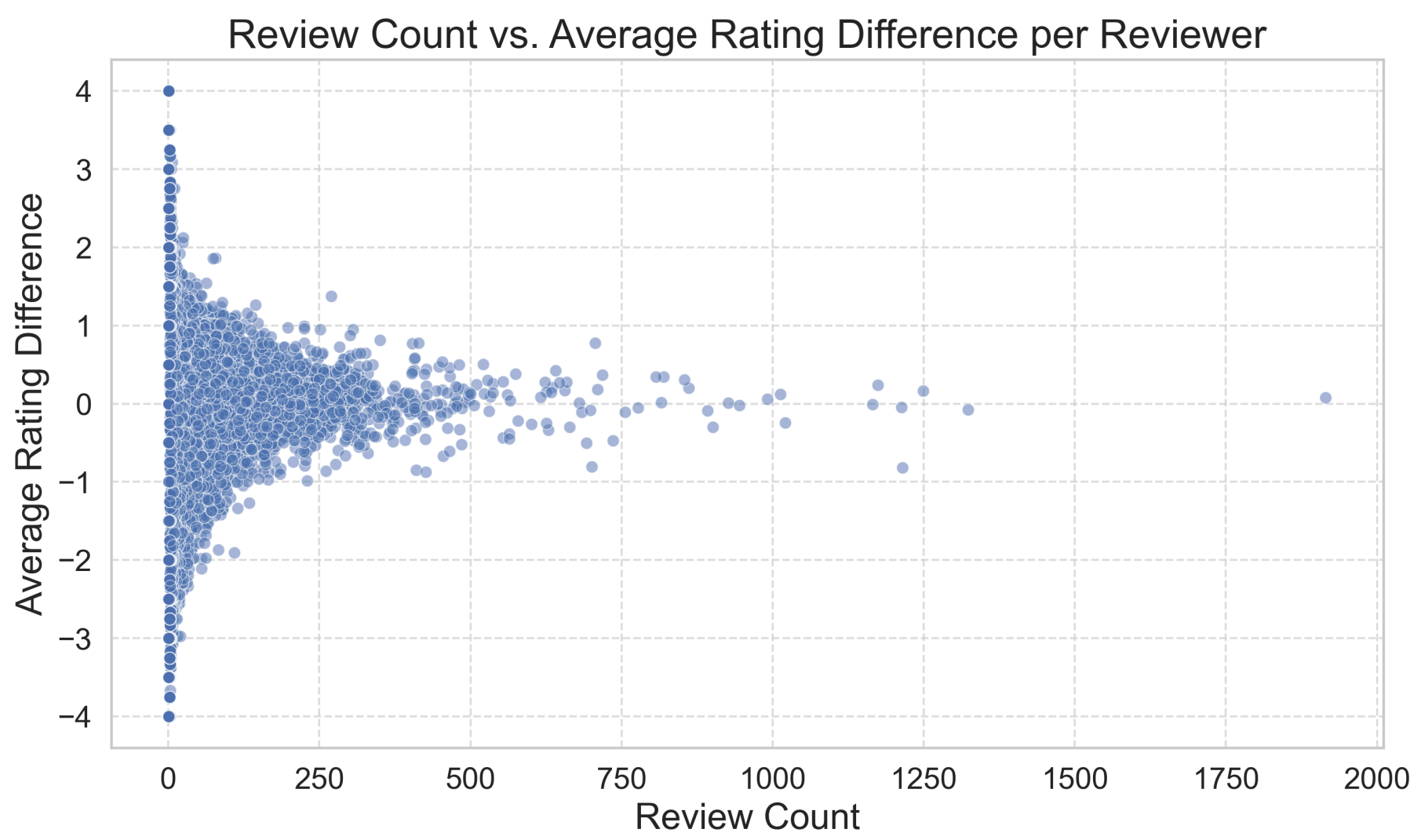
Adjustable features	Additional information
Food sentiment	Average sentiment
Ambiance sentiment	Parking
Service sentiment	Location
Value sentiment	Open/Close
Stars	Weekday/ Weekend
Review boost	rating
Category	Star Volatility

$$\text{Raw_Score} = \vec{x} \cdot \vec{w} \cdot \left(1 - \alpha \cdot \frac{v_s + v_r}{2}\right) \cdot r$$
$$\text{Score} = \frac{10 \cdot \text{Raw_Score}}{\max(\text{Raw_Score})}$$

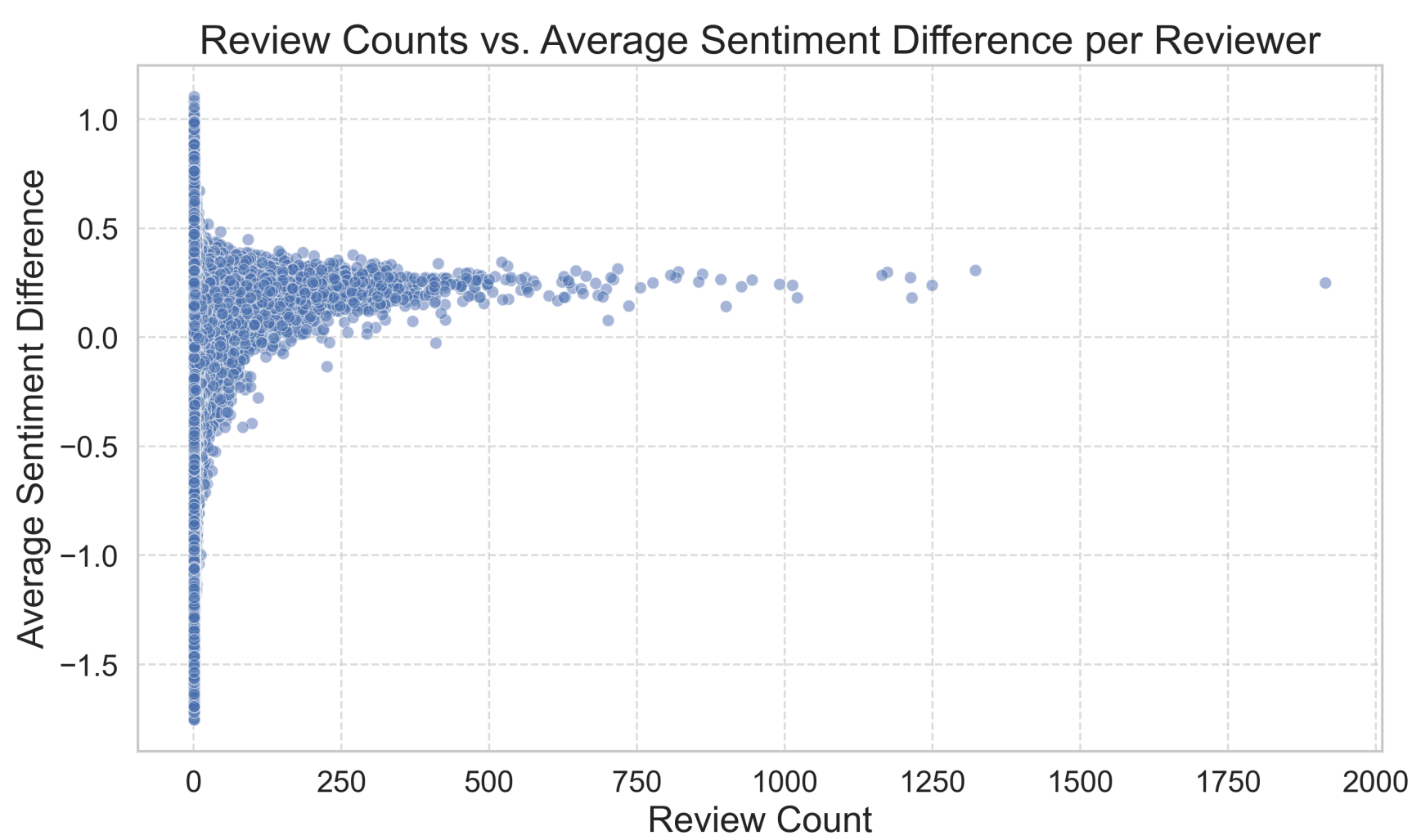
- $\vec{x} = [x_1, x_2, \dots, x_n]$: feature vector
- $\vec{w} = [w_1, w_2, \dots, w_n]$: normalized user-defined weights
- v_s : sentiment variability, v_r : star variability
- α : consistency penalty strength (e.g., 0.15)
- r : review confidence weight (based on review count)

3. Exploratory Data Analysis (EDA)

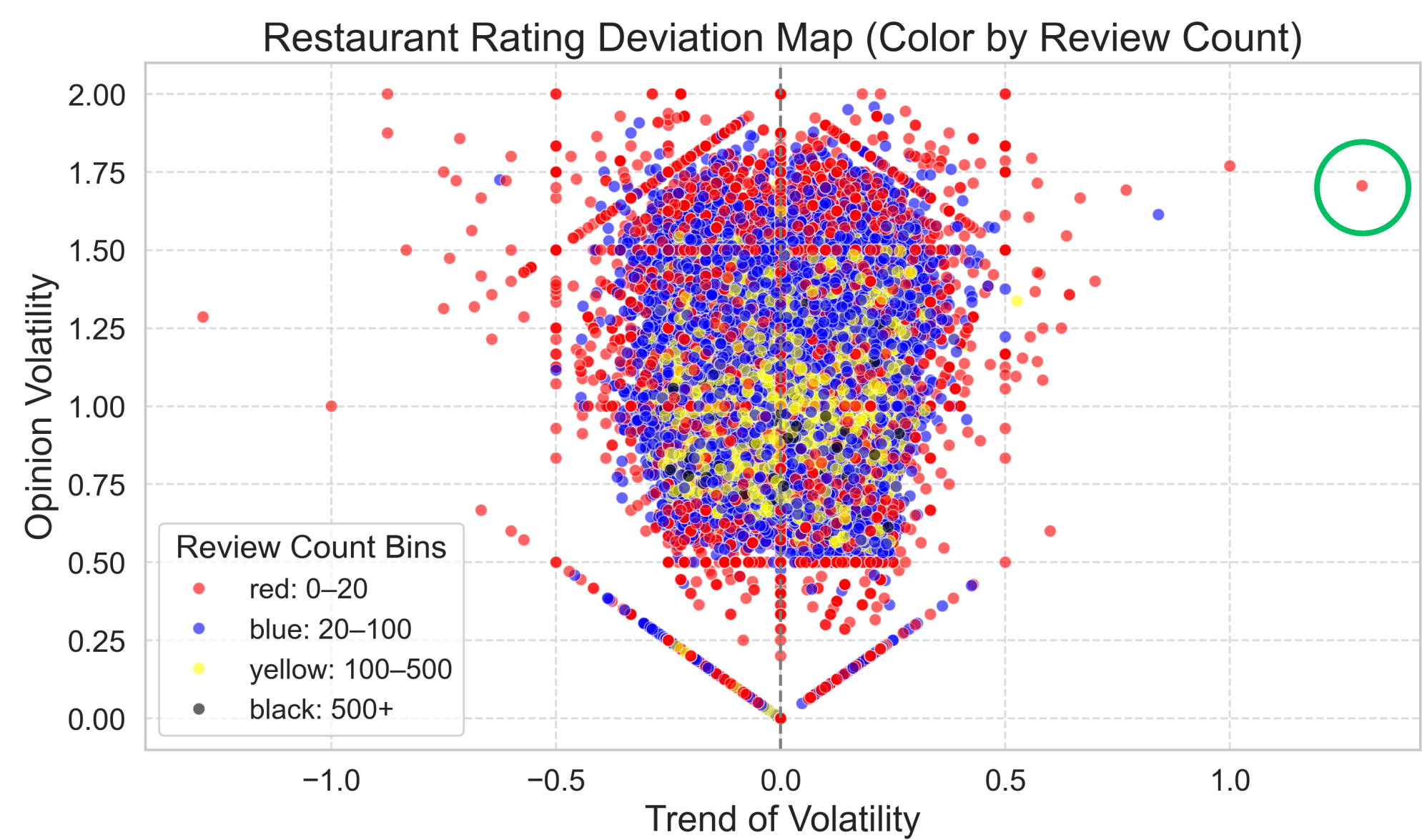
A Close-Up of Reviewers' Rating Consistency



The left graph shows individual reviewer behaviors, particularly those who give extreme ratings. As review count increases, rating deviations shrink and converge toward a mean difference of zero. We retain users with low review volume to preserve information, and later we penalize restaurants with high rating volatility.



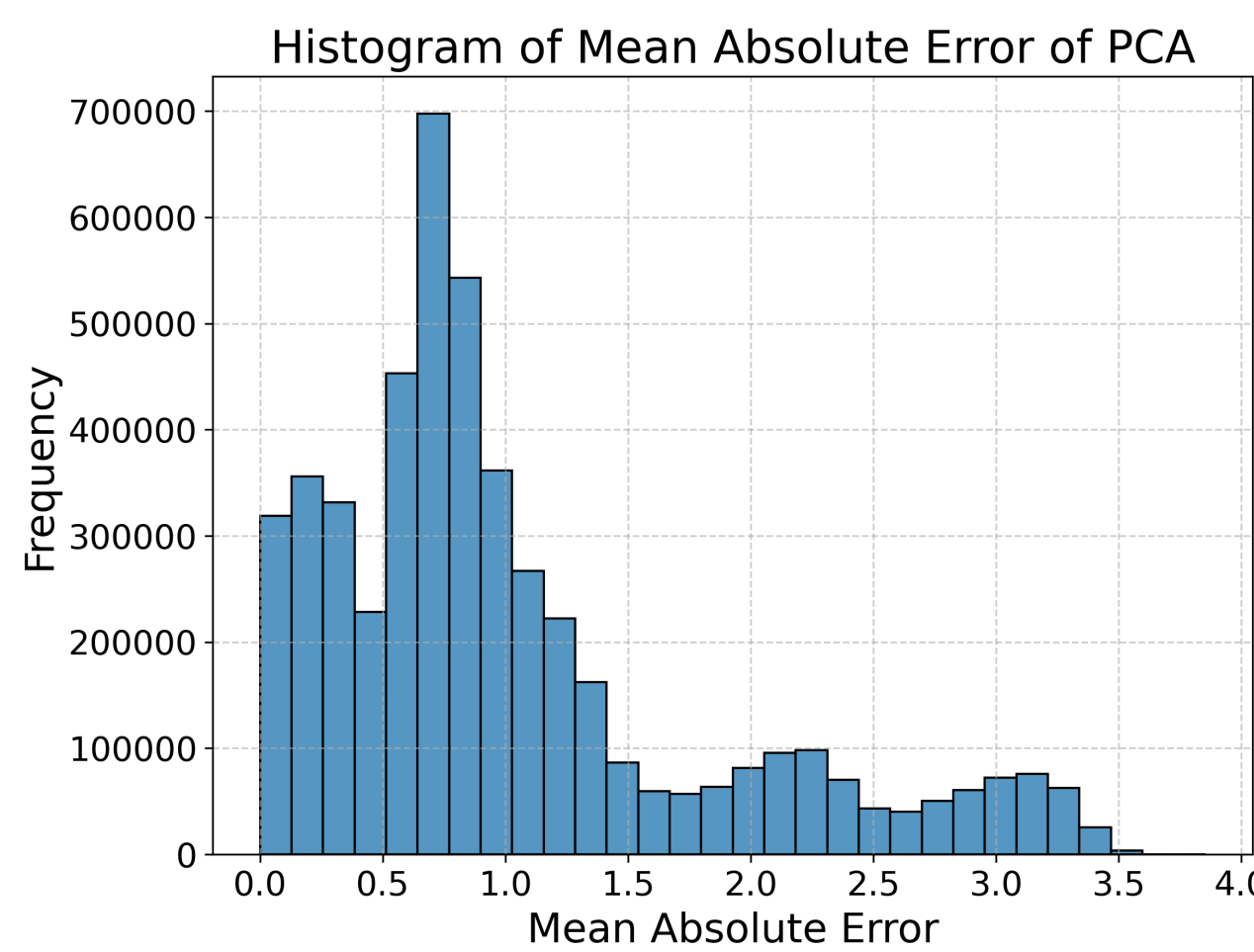
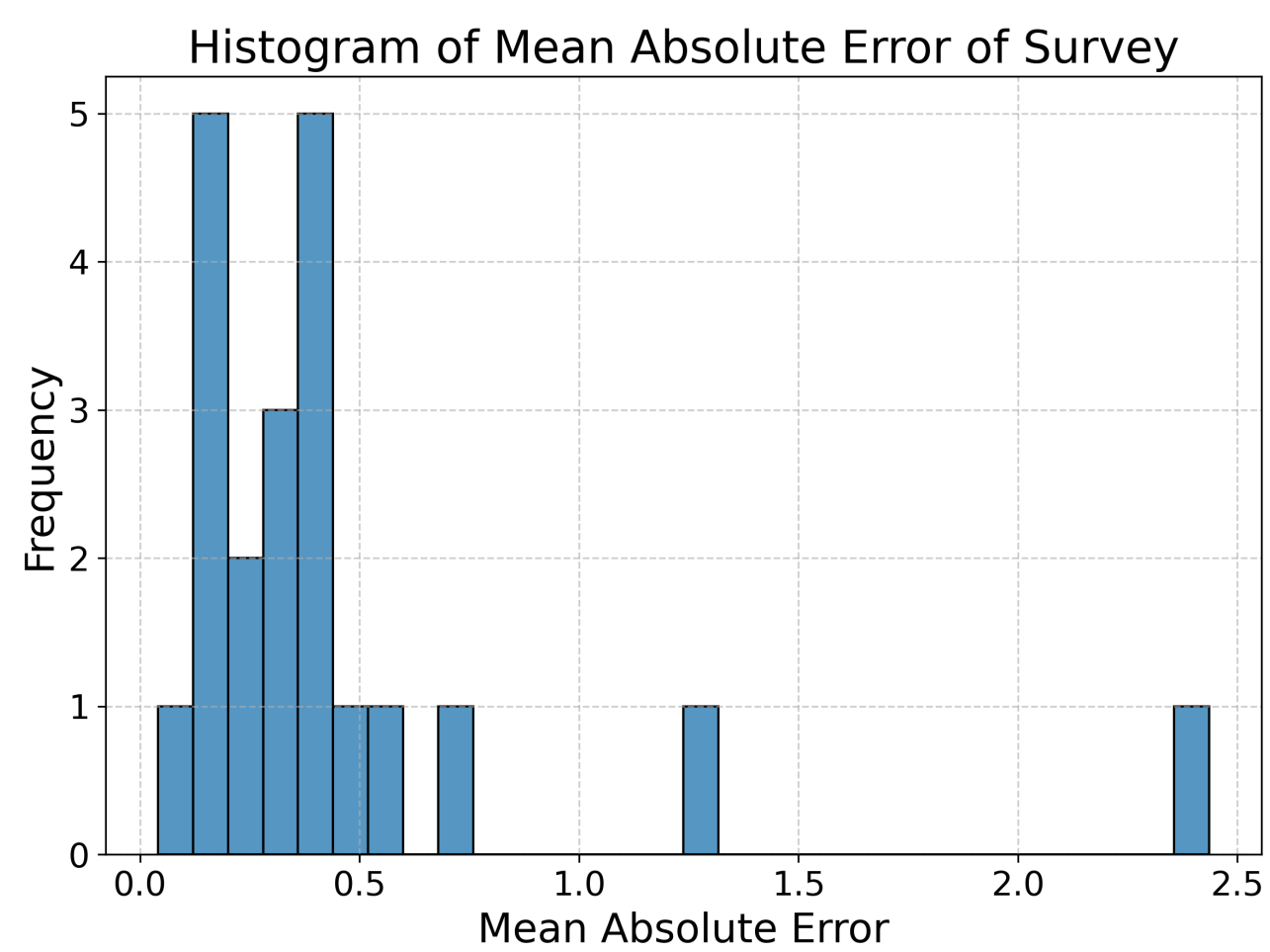
The right graph reveals an imperfection in our sentiment analysis. VADER, while efficient for general sentiment tasks, struggles with interpreting complex or nuanced texts, particularly sarcasm. This limitation likely contributes to the observed positive bias, as sarcastic comments may be misclassified as positive. We chose to keep users with few reviews to avoid losing information, and their stronger negative ratings help balance out the overall positive trend.



The final graph shows the volatility of each restaurant's ratings and its volatility trend. For example, the green-marked restaurant has high variability in opinions but a generally positive experience. This suggests it may be a risky choice, suitable for those seeking a more adventurous dining experience.

4. Recommendation Results and Conclusion

- A survey (n = 20) was conducted to assess our recommendation system by comparing it with the restaurant's average rating.
- Validation suggests a slight gain in performance when tailored to user preferences, though the small sample limits statistical confidence. (Model error = 0.94, baseline error = 1.02, t = -0.26, p = 0.79).
- In contrast, the default preferences weights (from PCA) comparison is more robust; the default setting is marginally lower compared to the base model. (PCA error = 2.00, baseline error = 1.98, t = 72, p < 0.05).



Limitations

- Outdated data results in limited and less reliable validation data (2015 dataset).
- Limited validation data hampers the ability to draw conclusive results.
- Difficulty in analyzing complex reviews introduces systematic errors in the model.

Next Steps

- Use alternative language models that identify sentiments more precisely
- Invest in a data scraper to get more recent and richer data instead of the open dataset
- Include proximity calculations in the user interface